

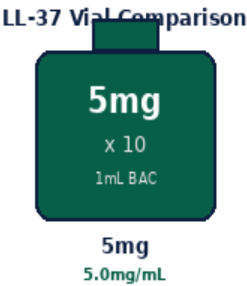
# LL-37

Human Cathelicidin | 5mg x 10 Vials

Antimicrobial Peptide | Immune Modulation | Wound Healing

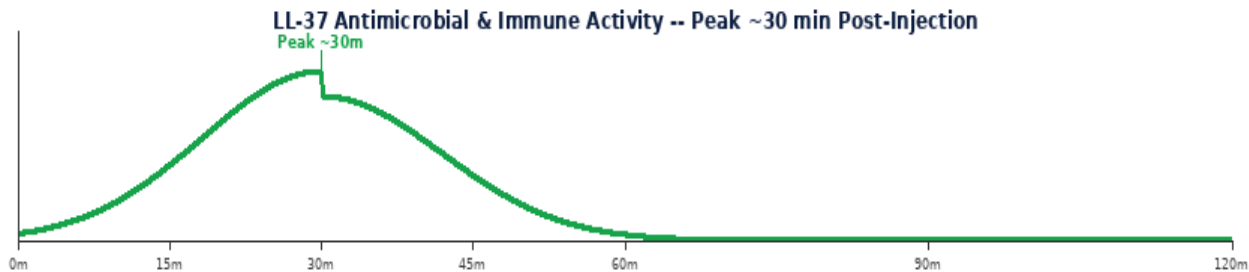
Purity: 99.8% | Endotoxin-Free | Research Use Only

LL-37 (also known as hCAP18 C-terminal peptide) is the only known human cathelicidin antimicrobial peptide. It is produced by neutrophils, macrophages, mast cells, and epithelial cells in response to infection and inflammation. LL-37 has broad-spectrum antimicrobial activity (bacteria, viruses, fungi), potent immune-modulatory effects, and promotes wound healing and angiogenesis. Injectable LL-37 is used in immunology, wound healing, and infection research.



| Parameter       | Specification                                                                     |
|-----------------|-----------------------------------------------------------------------------------|
| Class           | Human cathelicidin antimicrobial peptide (AMP) -- hCAP18 C-terminal               |
| Vial Size       | 5mg x 10 vials   Purity 99.8%                                                     |
| Reconstitution  | 1 mL BAC Water -> 5mg/mL (5,000 mcg/mL)                                           |
| Standard Dose   | 100-500 mcg per injection SubQ                                                    |
| Frequency       | 1-2x daily SubQ   no fasting required                                             |
| Key Effects     | Antimicrobial, antiviral, antifungal, immunomodulatory, wound healing, angiogenic |
| Cycle & Storage | 4-8 weeks on, 2-4 weeks off   2-8 deg C   use within 28 days                      |

MECHANISM OF ACTION & RESEARCH BENEFITS



| Pathway                      | Mechanism                                                                                                                                                                                                                            | Benefit                                                                      |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Antimicrobial Activity       | Disrupts bacterial membranes via electrostatic interaction with anionic phospholipids; active against gram-positive and gram-negative bacteria, enveloped viruses, and fungi; low resistance development vs conventional antibiotics | Infection research; antimicrobial resistance models; innate immunity studies |
| Immune Modulation            | Recruits neutrophils, monocytes, and mast cells to infection sites; promotes Th1/Th2 balance; modulates TLR signalling; anti-inflammatory at low concentrations; pro-inflammatory at high concentrations (dose-dependent)            | Innate immunity; inflammatory disease; autoimmune research                   |
| Wound Healing & Angiogenesis | Promotes keratinocyte migration and proliferation; stimulates angiogenesis via VEGF; accelerates re-epithelialisation; reduces biofilm formation; promotes tissue regeneration                                                       | Wound healing; chronic wound models; dermatological research                 |

| Benefit             | Context                                                                                                                                    |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Broad Antimicrobial | LL-37 has activity against bacteria, viruses, and fungi with low resistance development -- valuable for antimicrobial resistance research. |
| Immunomodulation    | Dose-dependent immune effects: anti-inflammatory at physiological concentrations, pro-inflammatory at high levels. Unique research tool.   |
| Wound & Skin        | Promotes keratinocyte migration, angiogenesis, and re-epithelialisation. Research applications in chronic wound and skin repair models.    |

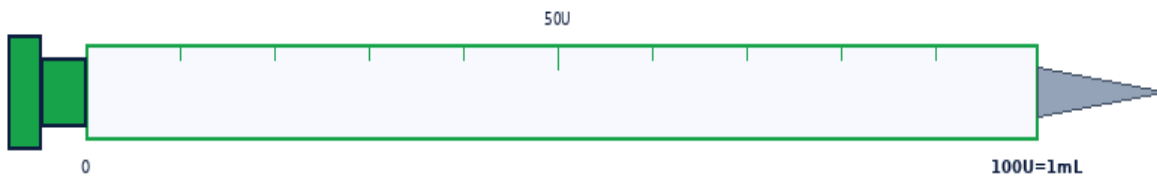
## RECONSTITUTION & DOSE CALCULATION

All vials use 1 mL BAC Water. Room temperature. Swirl gently -- never shake.



1 mL BAC Water per vial. Roll gently -- NEVER shake.

| Step | Action           | Detail                                                                                       |
|------|------------------|----------------------------------------------------------------------------------------------|
| 1    | Gather           | Vial, BAC Water, insulin syringe (29-31G), swabs, sharps.                                    |
| 2    | Clean stoppers   | Wipe both stoppers. Air-dry 10-15 sec.                                                       |
| 3    | Draw 1mL BAC     | Exactly 1 mL. Expel air bubbles.                                                             |
| 4    | Inject into vial | Aim at GLASS WALL. Slowly. Never onto powder.                                                |
| 5    | Swirl & label    | Roll 20-30 sec. Clear within 60 sec. Never shake. Label date. 2-8 deg C. Use within 28 days. |



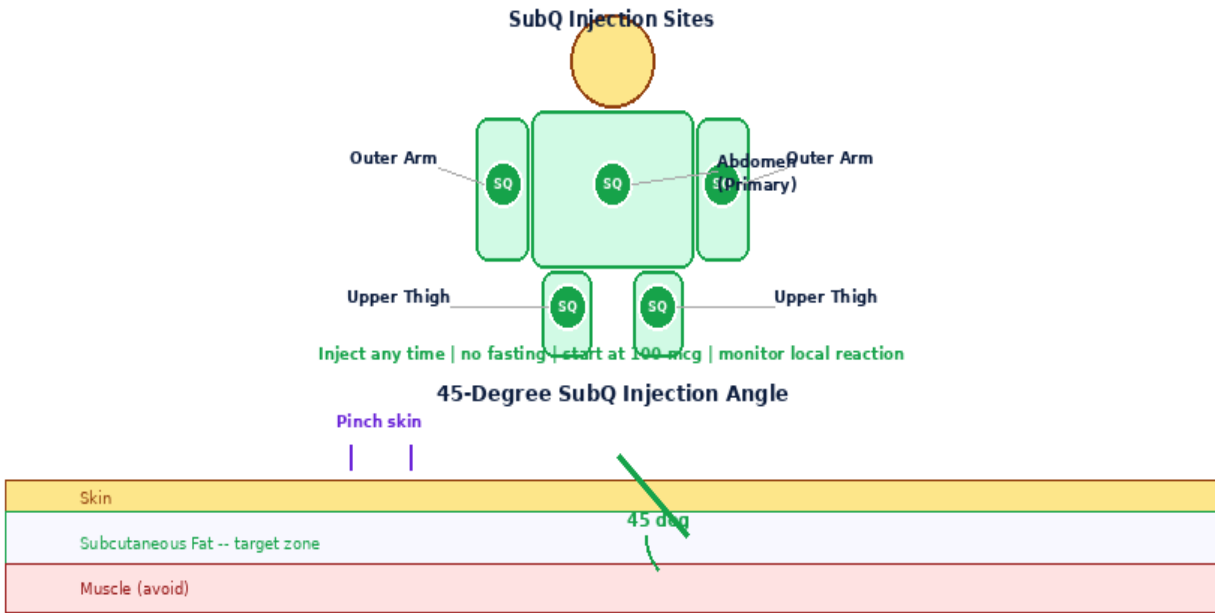
1mL Insulin Syringe | 29-31G | SubQ

**Formula:** Vol(mL)=Dose(mcg)/Conc(mcg/mL) | **Units:** Vol x 100

**5mg x 10 -- 1mL BAC -> 5.0mg/mL (5,000mcg/mL)**

| Dose(mcg)  | 100mcg | 200mcg | 300mcg | 400mcg | 500mcg |
|------------|--------|--------|--------|--------|--------|
| Vol(mL)    | 0.0200 | 0.0400 | 0.0600 | 0.0800 | 0.1000 |
| Units      | 2.0U   | 4.0U   | 6.0U   | 8.0U   | 10.0U  |
| Doses/Vial | 50     | 25     | 16     | 12     | 10     |

## INJECTION TECHNIQUE, PROTOCOLS & GOLDEN RULES



**8-Point Rotation Map -- Rotate Every Injection**



| Step | Action           | Detail                                           |
|------|------------------|--------------------------------------------------|
| 1    | Wash & prepare   | Wash hands. 29-31G syringe. No fasting required. |
| 2    | Draw dose        | Draw exact volume per dose table.                |
| 3    | Inject SubQ      | Abdomen. Pinch 2-3cm. 45 deg. Inject slowly.     |
| 4    | Rotate & dispose | 8-point rotation. Press 5 sec. Sharps container. |

| Protocol       | Dose        | Frequency  | Duration       | Notes                                     |
|----------------|-------------|------------|----------------|-------------------------------------------|
| Immune Support | 100-200 mcg | 1-2x daily | 4-8 wk 2-4 off | Lower dose. Anti-inflammatory range.      |
| Wound Healing  | 300-500 mcg | 1-2x daily | 4-8 wk         | Higher dose for wound and skin protocols. |

| Side Effect                   | Likelihood     | Management                                                    |
|-------------------------------|----------------|---------------------------------------------------------------|
| Injection site reaction       | Common         | Redness/stinging -- antimicrobial peptide. Rotate sites.      |
| Local inflammation            | Dose-dependent | Reduce dose if excess local reaction.                         |
| Pro-inflammatory at high dose | High dose      | LL-37 is pro-inflammatory above physiological range. Monitor. |

**Contraindications: Cancer history, pregnancy, under 18.**

|                                                                                       |
|---------------------------------------------------------------------------------------|
| 1. 1 mL BAC Water -> 5mg/mL (5,000 mcg/mL).                                           |
| 2. No fasting required. Inject any time of day.                                       |
| 3. Start at 100 mcg to assess local reaction before escalating.                       |
| 4. LL-37 is anti-inflammatory at low doses and pro-inflammatory at high doses.        |
| 5. Inject SubQ for systemic effects. Wound site injection for local healing research. |
| 6. Rotate through 8-point abdomen map. Use fresh 29-31G needle each injection.        |
| 7. Cycle 4-8 weeks on, 2-4 weeks off. Store 2-8 deg C, use within 28 days.            |
| 8. Monitor for excessive local inflammation -- reduce dose if occurs.                 |
| 9. Not for use during pregnancy, lactation, or in individuals under 18.               |

---

DISCLAIMER: SaiyanMed research use only. Not for human therapeutic use. Consult a qualified professional. SaiyanMed assumes no liability.

**SaiyanMed | [saiyanmed.com](https://saiyanmed.com) | Research Use Only**